Safe designs, unsafe ecosystems: deployment-layer selection in an evolutionary AI race

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Abstract

We study evolutionary dynamics in which the payoff earned in the game and the payoff that governs replication are different functions of the same interaction. The separation is the defining feature of an ecosystem of deployed AI agents: an agent acts in a competitive race, but whether its *design* is replicated is decided by a principal who does not play that race and who bears only a fraction λ of the external harm the design causes. We call the difference between the two functionals the selection wedge, embed it in the reduced four-strategy AI race of Fernández Domingos and Han, and analyse the resulting replicator and finite-population dynamics. The wedge collapses to a single control parameter, the effective liability $L = \lambda h$, and every invasion condition is affine in L, which yields closed-form thresholds. Four consequences follow. First, a conditionally unsafe design weakly dominates the unconditionally unsafe one at every L, so a pre-deployment filter that scores designs against a safe environment removes only a weakly dominated design and, under a common start measure, leaves the long-run unsafe frequency unchanged. Second, unconditional and conditional safety are exact payoff twins, so the invasion barrier of the safe face is set by a neutrally drifting composition variable, and it falls by a factor of 77.7 across that face. Third, long-run harm is *not monotone* in liability: on $L \in (4.27, 42.77)$ liability taxes the retaliation that conditional safety relies on, the unsafe coexistence becomes uninvadable, and crossing the lower edge from below lifts the basin-averaged unsafe frequency from 0 to 0.32. A window of this kind exists at every prize and private-risk level we examine, with multiplicative width between 4.2 and 12.8, and in a finite population the unsafe attractor survives between 65 and 3.9×10^{13} generations. Fourth, coupling the dynamics to a non-decreasing stock of diffused capability that erodes enforcement produces hysteresis whose width is exactly computable and independent of the speed of diffusion. The results locate the governance problem at the selection layer rather than at the design layer, and identify liability regimes that are worse than either extreme.

Keywords: evolutionary game theory; replicator dynamics; technological race; bistability; hysteresis; liability; AI safety

1 Introduction

Evolutionary game theory usually identifies two things that need not be the same: the payoff an individual earns in the game, and the quantity that governs how often that individual is copied [1–3]. Under that identification every standard result – the correspondence between rest points and Nash equilibria, the stability of evolutionarily stable strategies, the invasion analysis of finite populations – is a statement about one payoff matrix used twice.

Ecosystems of deployed AI agents break the identification in a specific way. An agent negotiates, bids, schedules or races on behalf of a principal, and earns a payoff by doing so. Whether the agent's *design* – its model, its system prompt, its scaffold – is replicated across further deployments is not decided by the agent, and is not decided by the payoff the agent earned in the abstract. It is decided by the principal who deployed it, on the basis of what the principal received. The two quantities differ whenever the design imposes costs on third parties that the principal does not bear. The replicating unit is therefore selected by an agent that never plays the game, according to a functional that is not the social one. Selection here is artificial rather than natural, in the breeder's sense: the copied variant is the one that served the breeder.

The consequences of that decoupling are the subject of this paper. They are not the familiar observation that individual incentives can diverge from collective welfare, which has been understood since Hardin's account of the commons [4] and is the content of every social dilemma. The point is structural and sharper: the dynamical system that describes the deployed population is the replicator equation of a game that nobody plays. Its rest points are the equilibria of the principals' selection game, and the social game enters only as a passive observable. Interventions therefore have very different effects depending on which of the two objects they touch. An intervention on the design pool – safety training, capability evaluations, release gating – changes the set of available strategies. An intervention on the selection functional – liability, disclosure, procurement rules – changes the vector field itself. We show that the first class can be exactly neutral while the second class is not, and that the second class can be non-monotone in a way that makes intermediate regimes worse than none.

The separation itself is not new. The indirect evolutionary approach [5–8] studies populations in which behaviour follows subjective preferences while selection operates on material payoffs, and Heifetz et al. [9] show that in almost every game some distortion of a player's true payoff is beneficial and survives payoff-monotone selection. Our construction is the mirror image. There the distortion sits in the behaviour and is selected on the socially correct functional, and it is endogenous; here behaviour follows the material race payoff, the distortion sits in the *selection* functional, and it is an exogenous policy parameter. What is new is not the existence of two functionals but the direction of the wedge and its consequences for governance instruments, which is why our results are statements about interventions rather than about which preferences survive.

Concern about selection pressure over deployed AI systems has been raised informally [10] and appears as one of seven risk factors in a recent multi-agent risk taxonomy [11], but we are not aware of a formal treatment. Formal analyses of AI races have instead studied the race itself: whether speed dominates safety [12], how regulation and heterogeneity change the prevalence of unsafe development [13, 14], and how commitments and delegation alter outcomes [15–17]. Those models keep the classical identification: the racing actors are the reproducing actors. We keep their interaction layer and change the selection layer.

Contribution. We formalise deployment-layer selection, embed it in the reduced AI-race game of Fernández Domingos and Han [18], and derive its consequences analytically and numerically. Section 2 specifies the two layers and the numerical protocol, Section 3 contains the analysis, and Section 4 draws out what the results imply for governance. The four results are:

1. **Solo evaluation cannot move the equilibrium** (Section 3.2). The conditionally antisocial design CAS weakly dominates the unconditionally unsafe design AU at every $L \geq 0$. A filter that rejects designs whose unsafe frequency against a safe environment exceeds ε rejects AU and admits CAS, whose self-play unsafe frequency is 1. The only attractor the filter deletes is one of maximal harm that the surviving pool reproduces.

2. **The safe face is neutral and its barrier is not** (Section 3.3). Always-safe and conditionally-safe designs are exact payoff twins, so the safe face carries no selection gradient. Its invasion barrier nonetheless falls from $L^* = 42.77$ at the unconditional vertex to $L^* = 0.55$ at the conditional vertex, a factor of 77.7; protection is governed by a variable that drifts.

3. **Long-run harm is non-monotone in liability** (Section 3.4). On $L \in (4.27, 42.77)$ the interior rest point of the unsafe face becomes uninvadable by conditional safety, because liability also taxes retaliation. Crossing the lower edge from below raises the basin-averaged unsafe frequency discontinuously from 0 to 0.32. The mixture-independence of the invasion criterion behind this result (Lemma 14) holds for any symmetric matrix game.

4. **Irreversible diffusion makes the transition hysteretic** (Section 3.7). With a monotone environmental state that erodes enforcement, the liability at which a safe regime is recovered exceeds the liability at which it was lost by a factor that is exactly computable and independent of the diffusion rate.

Three limitations should be stated before the results rather than after them. The interaction layer is the deliberately stylised two-player race of Fernández Domingos and Han [18] with four reduced strategies; it abstracts away many actors, gradual capability, and learning. The external harm h is a free parameter with no calibration, so we report the weight-free unsafe frequency as the primary observable, characterise the entire L-axis rather than committing to a point estimate, and normalise thresholds by the race prize. And the population is well mixed, so network reciprocity plays no role; we return to that in Section 4.

1.1 Related work

Evolutionary models of AI races. Armstrong et al. [12] introduced a formal race model in which information about rivals changes the willingness to skip safety precautions. Han et al. [13] embedded a race in an evolutionary dynamic and showed that whether regulation is beneficial depends on the speed advantage of unsafe development and on the time horizon; Cimpeanu et al. [14] extended the analysis to heterogeneous populations. These models establish that competitive pressure alone can sustain unsafe development. They also share the modelling choice we depart from: the racing actor and the reproducing actor coincide, so the population dynamic is driven by the racing payoff itself. Our contribution is orthogonal rather than corrective – we retain their race and ask what changes when the replication decision is made by a third party.

Delegation and artificial agents. Fernández Domingos et al. [16] showed experimentally that delegating a collective-risk decision to an artificial agent increases prosocial behaviour, and Terrucha et al. [17] showed that committing to an imperfect delegate can beat acting directly. That literature treats delegation as a commitment device chosen once by a player inside the game. Here the principal is outside the game and acts repeatedly on a population of designs; the object of interest is not the delegate's behaviour but the selection pressure the principal exerts on the design distribution.

Behavioural evidence on the race. The interaction layer we adopt is the framed experiment of Fernández Domingos and Han [18], in which paired participants repeatedly chose between safe and unsafe development under an uncertain horizon. Its central finding is that unsafe choices track the evolving strategic state of the race rather than elicited risk preferences. That finding motivates the conditional strategies CS and CAS, which carry all of our results. Conditional reciprocation of this kind is the dominant behavioural regularity in repeated social dilemmas more broadly [19–21].

Environmental feedback in evolutionary games. Coupling replicator dynamics to a slow environmental state produces oscillations and bistability [22, 23], and abrupt regime shifts with hysteresis are a well-studied feature of ecosystems with such feedbacks [24]. Existing game–environment models take the environment to be renewable: it recovers when exploitation stops. Deployed capability does not. Our ratchet is the degenerate but practically important case of a state that is monotone by construction, which converts a reversible parameter change into an irreversible regime change.

Liability. The economic analysis of accident law asks when strict liability outperforms negligence and when liability should be combined with regulation [25, 26]. Both analyses are equilibrium statements about a fixed population of injurers. We ask instead what a liability level does to an evolving distribution of designs. The answer is a qualitatively different phenomenon: a window of intermediate liability that is worse than either bound, because the instrument taxes the retaliation on which conditional safety depends. That taxation of retaliation is a second-order free-rider problem [27, 28] in a liability setting.

2 Model

We now specify the two layers. The interaction layer is taken unchanged from the source study [18] so that our results are attributable to the selection layer alone; the selection layer is new.

2.1 Interaction layer: the reduced AI race

Two agents repeatedly choose $a \in \{S, U\}$, safe or unsafe development. Safe development advances the race by $\sigma(S) = 1$ step and unsafe development by $\sigma(U) = 1.5$ steps. The round payoff matrix, rows indexed by the focal action, is

$$\pi = \begin{pmatrix} 1 & 0.6 \\ 2.4 & 2 \end{pmatrix}. \tag{1}$$

The race lasts at least five rounds and each further round ends it with probability 0.2, so the horizon is $T = 5 + G$ with $G \sim \mathrm{Geom}(0.2) - 1$ and $\mathbb{E}[T] = 9$. The agent with greater cumulative progress receives a prize B; equal progress splits it. A winning or tying agent suffers a setback with probability

$$q_i(T) = p_r^{\max} \frac{n_i^U(T)}{T}, \tag{2}$$

where $n_i^U(T)$ counts its unsafe choices, and a setback removes its task payoff. The baseline uses the experimental values $B = 100$ and $p_r^{\max} = 0.6$; Section 3.6 varies both, and Appendix B records the other two experimental treatments.

The reduced strategy set is the one inferred from the experiment: AS always safe; AU always unsafe; CS safe in round one, then copying the opponent's previous action; CAS unsafe in round one, then copying.

Remark 1 (Exact evaluation)**.** All four designs are deterministic, so the action path of an ordered pair is fixed once the pair is fixed and the only stochastic primitive is T. We therefore obtain the matchup quantities by exact expectation over the horizon law rather than by the Monte Carlo procedure used in the source study [18]. Exactness matters here because several of our thresholds are ratios of small payoff differences, so sampling noise of order 10^{-2} in the payoff matrix would propagate to the third significant digit of those ratios.

With the evaluation of Remark 1 in hand, write $a(i,j)$ for the expected task payoff of design i against j, $m(i,j)$ for its expected number of unsafe actions, and $u(i,j) = \mathbb{E}[n^U/T]$ for its expected unsafe frequency. Table 2 reports a and Table 3 reports m; both are reproduced with the threshold table in Appendix A.

2.2 Deployment layer: three functionals and the wedge

Each unsafe action imposes an external harm $h \geq 0$ on parties outside the race. Let $\lambda \in [0,1]$ be the fraction of that harm borne by the principal who deployed the design – the liability pass-through. That single parameter splits the interaction into three functionals:

$$\text{task payoff:} \quad a(i,j), \tag{3}$$

$$\text{selection functional:} \quad \pi_P(i,j) = a(i,j) - \lambda h\, m(i,j), \tag{4}$$

$$\text{social functional:} \quad \pi_S(i,j) = a(i,j) - h\, m(i,j). \tag{5}$$

Only π_P enters the dynamics. The social functional (5) is an accounting device: since each agent's harm is attributed to itself, the population mean of π_S equals mean task payoff minus total external harm.

Definition 2 (Selection wedge)**.** The selection wedge is $\Delta(i,j) = \pi_P(i,j) - \pi_S(i,j) = (1-\lambda)h\, m(i,j)$.

Figure 1 summarises the construction, and Figure 2 shows the three functionals and the wedge for the baseline race. The wedge is the entire formal content of the separation between the layer at which safety is engineered and the layer at which designs are selected.

2.3 Dynamics

Let $x \in \Delta_n$ be the frequency distribution over the n available designs. In an infinite population the deployment dynamics is the replicator equation of the selection functional,

$$\dot{x}_i \;=\; x_i\Big[(\pi_P x)_i - x^\top \pi_P x\Big], \qquad i = 1, \ldots, n. \tag{6}$$

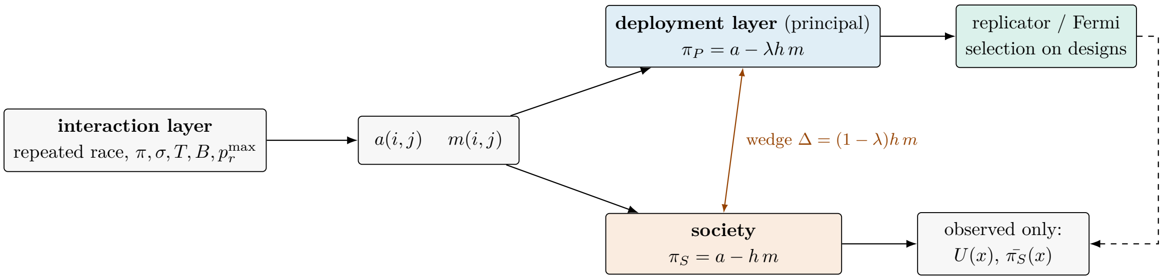

Figure 1: Deployment-layer selection. The interaction layer produces a task payoff a and an externality intensity m. The principal selects designs on π_P, which internalises a fraction λ of the harm; society is scored by π_S, which internalises all of it. Only the upper path feeds back into the population dynamics, so the social functional and the unsafe frequency $U(x)$ are observables of a system that does not optimise them. Their difference is the selection wedge Δ.

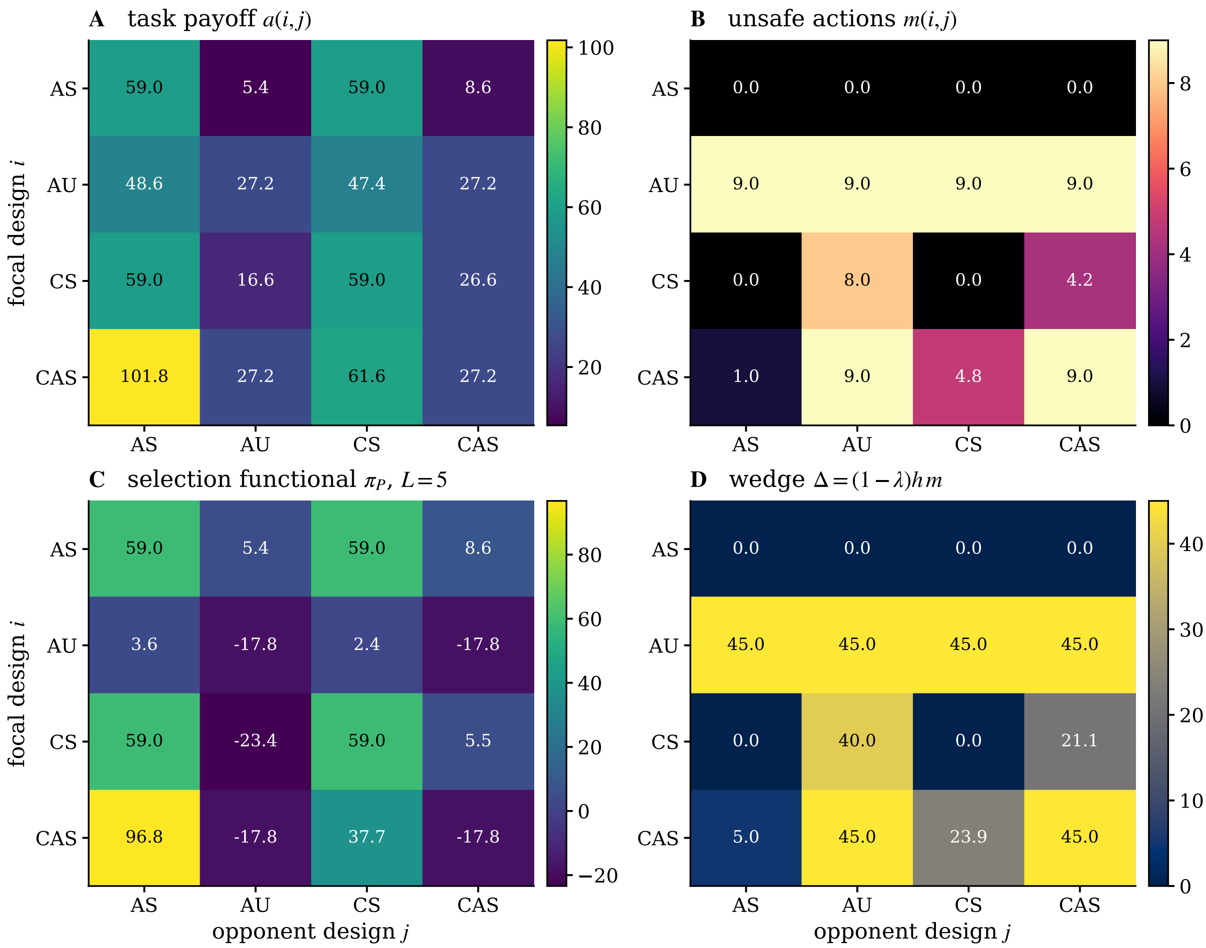

Figure 2: The three functionals of the baseline race ($B = 100$, $p_r^{\max} = 0.6$), rows indexed by the focal design. **(A)** Task payoff $a(i,j)$. **(B)** Expected number of unsafe actions $m(i,j)$. **(C)** Selection functional $\pi_P = a - Lm$ at effective liability $L = 5$. **(D)** The wedge $\Delta = (1-\lambda)hm$ at $\lambda = 0.5$, $h = 10$; it varies down the columns, which by Lemma 4 is exactly why it deforms the dynamics.

In a finite population of size Z we use the pairwise-comparison process with Fermi imitation [29]: a design B adopts A with probability $\left(1 + \exp[-\beta(f_A - f_B)]\right)^{-1}$, where β is the intensity of selection and f is computed from π_P, with mutation probability μ. We report both the full chain and its small-mutation limit [30]. All computations use EGTtools [31].

The primary observable is the population unsafe frequency

$$U(x) = \sum_{i,j} x_i x_j \, u(i,j), \tag{7}$$

which requires no welfare weights and is therefore independent of the uncalibrated harm parameter.

2.4 Numerical methods

Replicator trajectories are integrated with an adaptive LSODA solver at relative tolerance 10^{-10} and absolute tolerance 10^{-12}, to $t = 2\,000$, which is two orders of magnitude beyond the slowest relaxation we observe. Basin averages are taken over initial conditions drawn uniformly from the simplex, with 110 draws per parameter value in Figure 5 and 100 in Figure 3, from a fixed seed; doubling the number of draws changes the reported averages by less than 0.01. Where pools of different size are compared (Figure 3) initial conditions are drawn on the common $\{\mathsf{AS}, \mathsf{CS}, \mathsf{CAS}\}$ simplex and the extra design is seeded at a fixed share, so that the comparison is between one ecosystem with and without that design rather than between two different sampling measures; the reported curves use a 5% seed, and Section 3.6 reports the sensitivity to that choice. Finite-population stationary distributions are obtained by solving the singular linear system for the stationary vector of the sparse transition matrix, since densifying it is infeasible at the state-space sizes used here. Every number quoted in the text is regenerated by the released code.

3 Results

3.1 Reduction, inertness and invasion thresholds

This section assembles the analytical scaffolding. The first two statements are immediate or classical and are recalled rather than claimed; the closed forms in Lemma 7 and the general criterion in Lemma 14 do the work later.

Proposition 3 (Reduction to the effective liability)**.** *The dynamics* (6) *and the Fermi process depend on* (λ, h) *only through the product* $L = \lambda h$.

Proof. Both dynamics depend on π_P only, and by (4) $\pi_P = a - L\,m$. □

We call L the *effective liability*, measured in payoff units per unsafe action. It is the only policy variable in what follows; a regulator can move it by raising the pass-through λ at fixed harm, and the same outcome obtains for any (λ, h) on the hyperbola $\lambda h = L$. Because L inherits the scale of the race, we also report L/B.

Lemma 4 (Column-constant invariance; classical)**.** *Let* $A' = A + \mathbf{1}c^\top$ *for some* $c \in \mathbb{R}^n$*. Then* A *and* A' *induce the same replicator field on* Δ_n *and have the same symmetric Nash set.*

Proof. $(A'x)_i = (Ax)_i + c^\top x$ for every i, so all fitnesses shift by the same state-dependent constant and the differences $(A'x)_i - x^\top A'x$ are unchanged; likewise every pairwise payoff comparison at a fixed opponent distribution is unchanged [3]. □

Definition 5. A wedge Δ is *strategically inert* if $\Delta = \mathbf{1}c^\top$, that is, if $\Delta(i,j)$ does not depend on the focal design i.

Proposition 6 (Inertness only at full pass-through)**.** *In the reduced race game, $\Delta = (1-\lambda)h\, m$ is strategically inert if and only if $\lambda = 1$ or $h = 0$. The deployment dynamics is therefore orbit-equivalent to the dynamics of the social functional exactly at full liability pass-through.*

Proof. By Lemma 4 inertness holds iff $m(i,j)$ is independent of i or the prefactor vanishes. Table 3 gives $m(\mathsf{AS},\mathsf{AS}) = 0 \neq 9 = m(\mathsf{AU},\mathsf{AS})$, so m depends on the focal design and inertness requires $(1-\lambda)h = 0$. □

Proposition 6 has a practical reading: there is no cheap partial liability that reproduces the socially correct dynamics. Any $\lambda < 1$ deforms the vector field, and the deformation is proportional to $1-\lambda$. It does not follow that the deformation is harmful in proportion to its size; Section 3.4 shows the opposite over a wide range of L. The proposition also identifies, by contraposition, the one liability rule that would be dynamically harmless: a rule whose charge depends on the counterparty's behaviour rather than on one's own is column-constant and, by Lemma 4, leaves the replicator field untouched.

Lemma 7 (Affine invasion conditions)**.** *A rare design i in a resident population of j has positive growth rate under* (6) *if and only if*

$$\delta a - L\,\delta m > 0, \qquad \delta a := a(i,j) - a(j,j), \quad \delta m := m(i,j) - m(j,j). \tag{8}$$

If $\delta m \neq 0$ the critical effective liability is $L^* = \delta a/\delta m$; the invader is favoured for $L < L^*$ when $\delta m > 0$ and for $L > L^*$ when $\delta m < 0$. If $\delta m = 0$ the outcome does not depend on L.

Proof. The growth rate of a rare i is $\pi_P(i,j) - \pi_P(j,j)$, which by (4) equals $\delta a - L\,\delta m$. The case distinction follows by solving a linear inequality. □

Applying Lemma 7 to every ordered pair gives Table 4. Four of the twelve thresholds organise the rest of the paper:

$$\begin{aligned} &\underbrace{L^*_{\mathsf{CS}\to\mathsf{CAS}} = 0.116}_{\text{conditional safety enters}} \;<\; \underbrace{L^*_{\mathsf{CAS}\to\mathsf{CS}} = 0.551}_{\text{conditional safety protected}} \\ &\quad <\; \underbrace{L^\dagger = 4.274}_{\text{retaliation becomes too costly}} \;<\; \underbrace{L^*_{\mathsf{CAS}\to\mathsf{AS}} = 42.765}_{\text{unconditional safety protected}}\;. \end{aligned} \tag{9}$$

The spread of (9) over a factor of 368 is the quantitative signature of the results below. In units of the prize the outer thresholds are $L^*_{\mathsf{CAS}\to\mathsf{CS}}/B = 0.0055$ and $L^*_{\mathsf{CAS}\to\mathsf{AS}}/B = 0.428$.

3.2 Solo evaluation cannot move the equilibrium

Pre-deployment safety evaluation probes a model in isolation and admits it when its behaviour is acceptable. We formalise this as a filter on the design pool and show that it is dynamically inert.

Definition 8 (Solo-evaluation filter)**.** The solo score of design i is $s(i) = u(i,\mathsf{AS})$, its unsafe frequency against a fully safe environment. The filter of strictness ε admits $\Theta_\varepsilon = \{i : s(i) \le \varepsilon\}$.

In the reduced game, $s(\mathsf{AS}) = s(\mathsf{CS}) = 0$, $s(\mathsf{AU}) = 1$, and CAS defects exactly once before its opponent's safe play locks it into safe play, so

$$s(\mathsf{CAS}) = \mathbb{E}\Big[\tfrac{1}{T}\Big] = \sum_{g \geq 0} \frac{0.2 \cdot 0.8^g}{5 + g} = 0.132. \tag{10}$$

Any filter with $\varepsilon \in [0.132, 1)$ therefore admits $\{\mathsf{AS}, \mathsf{CS}, \mathsf{CAS}\}$ and rejects only AU. The next result says that the rejection is dynamically empty.

Theorem 9 (Weak dominance of CAS over AU)**.** *For every $L \geq 0$ and every opponent j, $\pi_P(\mathit{CAS}, j) \geq \pi_P(\mathit{AU}, j)$, with strict inequality for $j \in \{\mathit{AS}, \mathit{CS}\}$.*

Proof. From Tables 2 and 3,

$$\pi_P(\mathsf{CAS}, j) - \pi_P(\mathsf{AU}, j) = \begin{cases} 53.125 + 8L, & j = \mathsf{AS}, \\ 14.271 + 4.222L, & j = \mathsf{CS}, \\ 0, & j \in \{\mathsf{AU}, \mathsf{CAS}\}, \end{cases} \tag{11}$$

each of which is non-negative for $L \geq 0$ and strictly positive in the first two cases. Against AU and CAS the two designs generate identical action paths from round two onwards and identical unsafe counts, so the difference vanishes identically. □

The mechanism behind Theorem 9 is transparent: AU and CAS behave the same way against an unsafe environment, but CAS stops defecting against a safe one, which earns it more and costs it less. Unconditional aggression is dominated by conditional aggression at every liability level.

Corollary 10 (Filter futility)**.** *Let $\Theta = \{\mathit{AS}, \mathit{AU}, \mathit{CS}, \mathit{CAS}\}$ and $\varepsilon \in [0.132, 1)$, so that $\Theta_\varepsilon = \{\mathit{AS}, \mathit{CS}, \mathit{CAS}\}$. Then, for every $L \geq 0$:*

(i) along every interior solution of (6) the ratio $x_{\mathsf{AU}}/x_{\mathsf{CAS}}$ is non-increasing, and strictly decreasing whenever $x_{\mathsf{AS}} + x_{\mathsf{CS}} > 0$;

(ii) any attractor of Θ that the filter removes carries the maximal unsafe frequency $U = 1$, so the filter cannot delete an attractor with lower harm than one it leaves behind;

(iii) under a start measure common to both pools, the basin-averaged long-run unsafe frequencies of Θ and Θ_ε agree to within 1.2×10^{-3} across the whole L-axis.

Proof. For (i), (11) gives $\frac{d}{dt}\log(x_{\mathsf{AU}}/x_{\mathsf{CAS}}) = \pi_P(\mathsf{AU}, x) - \pi_P(\mathsf{CAS}, x) = -[(53.125 + 8L)x_{\mathsf{AS}} + (14.271 + 4.222L)x_{\mathsf{CS}}] \leq 0$. For (ii), by (i) an ω-limit with $x_{\mathsf{AU}} > 0$ must satisfy $x_{\mathsf{AS}} = x_{\mathsf{CS}} = 0$ and hence lie in the $\{\mathsf{AU}, \mathsf{CAS}\}$ edge, on which the two designs are payoff-identical, every point is a rest point, and $u \equiv 1$. Part (iii) is numerical and reported in Figure 3A. □

Part (ii) is weaker than one might hope, and the weakness is itself informative. Filtering AU does delete an attractor: for $L < L^*_{\mathsf{AS}\to\mathsf{AU}} = 2.422$ an AU-dominated population is neutrally stable, and removing AU removes it. But that attractor has $U = 1$, exactly the unsafe frequency of the CAS vertex that the filter leaves in place whenever $L < L^*_{\mathsf{CS}\to\mathsf{CAS}} = 0.116$. The filter changes which design occupies the all-unsafe state, not whether the state exists, and the state is reachable only from populations already dominated by AU.

Figure 3 makes the comparison on a common start measure. On that comparison the curves coincide over the whole L-axis. Panel B shows why. A solo evaluation sees $s(\mathsf{CAS}) = 0.132$ and cannot distinguish CAS from a safe design at any threshold that admits reciprocation at all, while CAS in self-play produces $u = 1$ – the same equilibrium harm as the design the filter did remove.

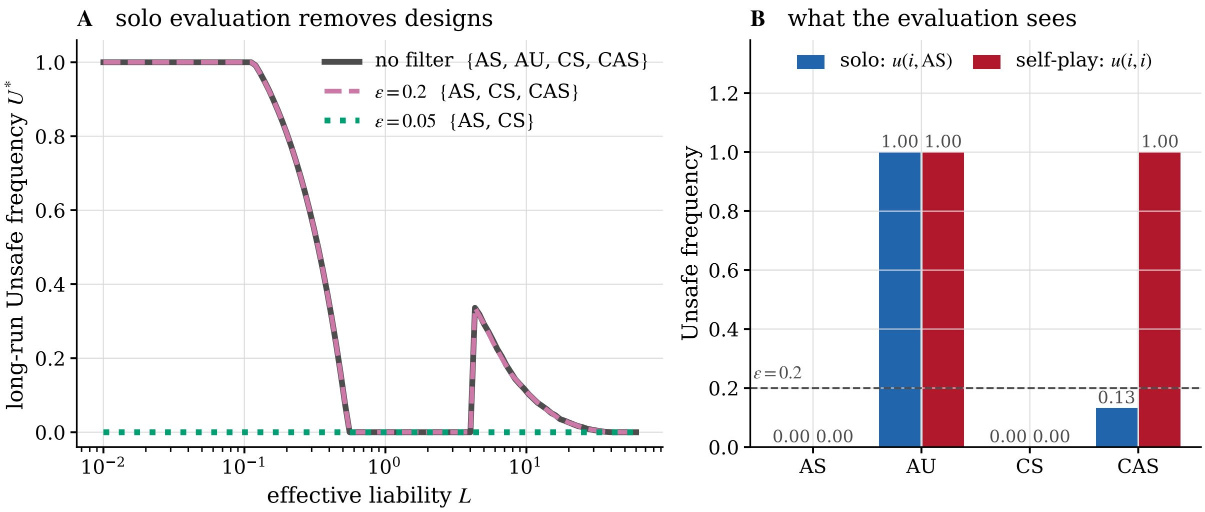

Figure 3: Solo evaluation filters are dynamically empty. **(A)** Long-run unsafe frequency of the unfiltered pool $\{\mathsf{AS}, \mathsf{AU}, \mathsf{CS}, \mathsf{CAS}\}$ and of the pools admitted at $\varepsilon = 0.2$ and $\varepsilon = 0.05$. Initial conditions are drawn on the $\{\mathsf{AS}, \mathsf{CS}, \mathsf{CAS}\}$ simplex for every pool, with AU seeded at a fixed 5% where it is present, so that the three curves describe the same ecosystem. The first two coincide everywhere, as Corollary 10(iii) requires. **(B)** What the evaluation sees. Blue: unsafe frequency against a safe environment, the quantity the filter measures; the AS and CS bars are zero. Red: unsafe frequency in self-play, the quantity that determines equilibrium harm. The two orderings disagree precisely for CAS.

A strict filter with $\varepsilon < 0.132$ does remove CAS, leaving the pool $\{\mathsf{AS}, \mathsf{CS}\}$ on which $U^* = 0$ for every L. Two caveats make this less reassuring than it looks. First, that pool is exactly the neutral safe face of Section 3.3, so its protection lasts only as long as no CAS-like design re-enters, and Theorem 12 bounds how much liability is needed when one does. Second, the strict threshold is only achievable because the probe environment in (10) is fully safe: measured against an unsafe opponent the conditionally safe design scores $u(\mathsf{CS}, \mathsf{CAS}) = 0.461$ and would be rejected by the same rule. A filter strict enough to exclude conditional aggression in a mixed environment also excludes conditional safety, which Theorem 12 identifies as the cheapest design to protect.

3.3 A neutral safe face with a drifting barrier

Proposition 11 (Payoff twins)**.** *Restricted to* $\{\mathsf{AS}, \mathsf{CS}\}$*, the selection functional is constant:* $\pi_P(i, j) = a(\mathsf{AS}, \mathsf{AS})$ *for all* $i, j \in \{\mathsf{AS}, \mathsf{CS}\}$ *and all* L*. The* AS–CS *edge is therefore a set of rest points of* (6)*, and in the finite-population process it evolves by neutral drift.*

Proof. A conditionally safe design plays safe in round one and then copies a safe opponent, so all four matchups within $\{\mathsf{AS}, \mathsf{CS}\}$ realise the all-safe path. Hence $a \equiv 59$, $m \equiv 0$ and $\pi_P \equiv 59$ independently of L. □

Neutrality does not make the composition irrelevant, because AS and CS differ against an invader.

Theorem 12 (Composition-dependent invasion barrier)**.** *Let the resident population be the mixture* $(1-x)\mathsf{AS} + x\mathsf{CS}$ *on the safe face, and write* $\delta a_k = a(\mathsf{CAS}, k) - a(\mathsf{AS}, \mathsf{AS})$ *and* $\delta m_k = m(\mathsf{CAS}, k)$ *for* $k \in \{\mathsf{AS}, \mathsf{CS}\}$*. A rare* CAS *invader has positive growth rate if and only if* $L < L^*(x)$*, where*

$$L^*(x) = \frac{(1-x)\,\delta a_{\mathsf{AS}} + x\,\delta a_{\mathsf{CS}}}{(1-x)\,\delta m_{\mathsf{AS}} + x\,\delta m_{\mathsf{CS}}}, \qquad \begin{array}{ll} \delta a_{\mathsf{AS}} = 42.765, & \delta m_{\mathsf{AS}} = 1, \\ \delta a_{\mathsf{CS}} = 2.631, & \delta m_{\mathsf{CS}} = 4.778. \end{array} \tag{12}$$

L^* *is strictly decreasing on* $[0, 1]$*, with* $L^*(0) = 42.765$ *and* $L^*(1) = 0.551$*, a ratio of* 77.7*.*

Proof. By Proposition 11 the resident fitness is $a(\mathsf{AS}, \mathsf{AS})$ for every x, so the invader's growth rate is $(1-x)(\delta a_{\mathsf{AS}} - L\delta m_{\mathsf{AS}}) + x(\delta a_{\mathsf{CS}} - L\delta m_{\mathsf{CS}})$, which is positive exactly when $L < L^*(x)$. Differentiating the ratio of the two affine functions, the sign of $\frac{d}{dx}L^*$ is that of $\delta a_{\mathsf{CS}}\delta m_{\mathsf{AS}} - \delta a_{\mathsf{AS}}\delta m_{\mathsf{CS}} = 2.631 - 204.34 < 0$. □

Inverting (12) gives the critical share of conditional safety $x^*(L)$ that a given liability level requires: $x^*(1) = 0.951$, $x^*(2) = 0.855$, $x^*(5) = 0.640$, $x^*(20) = 0.197$. A population that is 90% unconditionally safe is invadable at any $L < 28.1$, whereas the same population with the shares reversed is already protected at $L = 1.6$.

The combination of Proposition 11 and Theorem 12 is the substantive point. The variable that determines whether a safe population can be invaded carries no selection gradient of its own, so it is free to drift. Figure 4C shows the stationary distribution of the neutral edge in a finite population: for small mutation the mass concentrates at the two vertices, so the population spends long stretches near pure AS, which is the maximally vulnerable composition. Protection obtained by mandating unconditional safety is therefore not durable – not because the mandate is violated, but because the direction it protects against is selectively invisible.

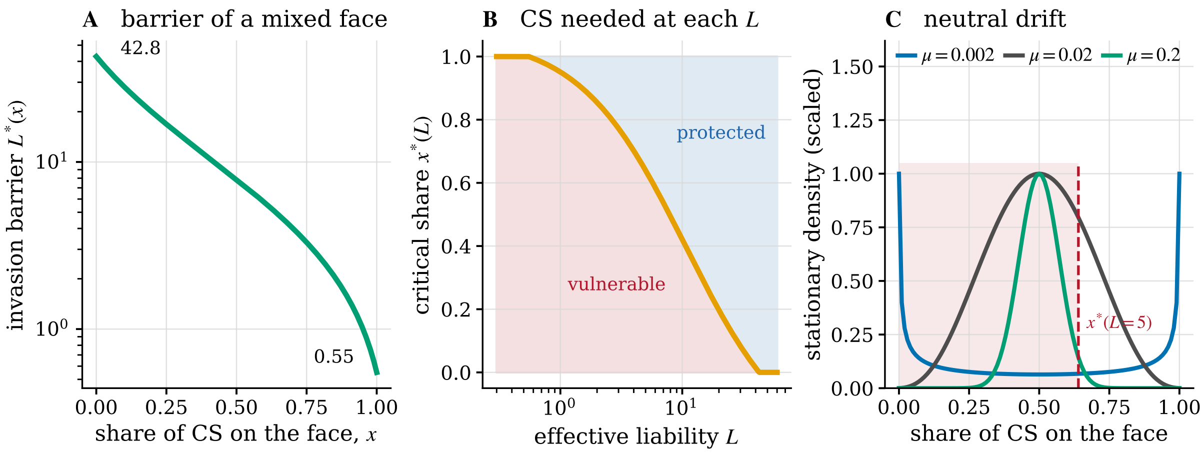

Figure 4: The safe face is neutral but not uniform. **(A)** Invasion barrier $L^*(x)$ of Theorem 12, log scale. **(B)** The critical share of conditional safety $x^*(L)$ separating vulnerable from protected compositions. **(C)** Stationary density of the neutral AS–CS edge in a finite population ($Z = 100$, $\beta = 0.05$) for three mutation rates, with the vulnerability threshold at $L = 5$ marked. Small mutation concentrates the mass at the vertices, half of it on the vulnerable side.

3.4 Bistability and the liability valley

The results so far might suggest that harm decreases with liability. It does not. This section identifies a window of intermediate liability on which the system is bistable and the basin-averaged harm jumps upward.

Proposition 13 (Interior rest point of the unsafe face)**.** *On the* $\{\mathsf{AS}, \mathsf{CAS}\}$ *face the replicator field has an interior rest point whenever* $\alpha := \pi_P(\mathsf{CAS}, \mathsf{AS}) - \pi_P(\mathsf{AS}, \mathsf{AS})$ *and* $\gamma := \pi_P(\mathsf{AS}, \mathsf{CAS}) - \pi_P(\mathsf{CAS}, \mathsf{CAS})$ *share a sign, located at* $p^* = \alpha/(\alpha+\gamma)$ *in the* CAS *direction, and it attracts the face when both are positive. With the values of Table 2,*

$$p^*(L) = \frac{42.765 - L}{24.165 + 8L}, \qquad L \in (L^*_{\mathsf{AS}\to\mathsf{CAS}},\, L^*_{\mathsf{CAS}\to\mathsf{AS}}) = (2.067,\, 42.765). \tag{13}$$

Proof. Direct computation of the two-strategy replicator rest point, followed by substitution of $\alpha = 42.765 - L$ and $\gamma = 9L - 18.6$. $\square$

The next statement is the mathematical core of the section, and it is not specific to the race game.

Lemma 14 (Mixture-free invasion at a twin face)**.** *Let* M *be a symmetric matrix game, let* i, j *span a face with an interior rest point at frequency* $p \in (0, 1)$ *of* j*, and let* k *be a third strategy that is a payoff twin of* i *against* i*, that is* $M_{ki} = M_{ii}$*. Then the growth rate of a rare* k *at that rest point equals* $p\,(M_{kj} - M_{ij})$*. Its sign is therefore independent of* p*, and* k *invades the interior rest point if and only if* $M_{kj} > M_{ij}$*.*

Proof. At an interior rest point of the face the two residents have equal fitness $f = (1-p)M_{ii} + pM_{ij}$. The growth rate of a rare k is $(1-p)M_{ki} + pM_{kj} - f = (1-p)(M_{ki} - M_{ii}) + p(M_{kj} - M_{ij})$, and the first term vanishes by hypothesis. $\square$

Lemma 14 says that whenever a candidate entrant is behaviourally indistinguishable from an incumbent against that incumbent, whether it can enter a mixed equilibrium involving the incumbent is decided by a single pairwise comparison, with no dependence on the equilibrium mixture. Applied to our game with $i = \mathsf{AS}$, $j = \mathsf{CAS}$, $k = \mathsf{CS}$ – the hypothesis is exactly Proposition 11 – it yields the threshold that governs the valley.

Theorem 15 (Guard threshold)**.** *CS invades the interior rest point of the* $\{\mathsf{AS}, \mathsf{CAS}\}$ *face if and only if*

$$L < L^\dagger := \frac{a(\mathsf{CS}, \mathsf{CAS}) - a(\mathsf{AS}, \mathsf{CAS})}{m(\mathsf{CS}, \mathsf{CAS}) - m(\mathsf{AS}, \mathsf{CAS})} = \frac{26.644 - 8.600}{4.222 - 0} = 4.274. \tag{14}$$

Proof. Apply Lemma 14 with $M = \pi_P$, $i = \mathsf{AS}$, $j = \mathsf{CAS}$, $k = \mathsf{CS}$; the twin hypothesis $\pi_P(\mathsf{CS}, \mathsf{AS}) = \pi_P(\mathsf{AS}, \mathsf{AS})$ is Proposition 11. The criterion $\pi_P(\mathsf{CS}, \mathsf{CAS}) > \pi_P(\mathsf{AS}, \mathsf{CAS})$ is affine in L by (4), and solving it gives (14). $\square$

The interpretation of $L^\dagger$ is the crux of the paper. A conditionally safe design defends a population by retaliating: it matches unsafe play, which is what makes exploiting it unprofitable. Retaliation is itself unsafe behaviour, so liability taxes it. Above $L^\dagger$ the tax on retaliation exceeds the tax the retaliation imposes on the aggressor, the guard loses its advantage, and the unsafe coexistence becomes uninvadable.

Corollary 16 (Liability valley)**.** *For $L \in (L^{\dagger}, L^{*}_{\mathsf{CAS}\to\mathsf{AS}}) = (4.274, 42.765)$ the dynamics on $\{\mathsf{AS}, \mathsf{CS}, \mathsf{CAS}\}$ has two attractors: the neutral safe face with $U = 0$, and the interior rest point (13) with $U = p^{*}(1-p^{*})u(\mathsf{CAS}, \mathsf{AS}) + (p^{*})^{2} > 0$. At the lower edge the unsafe attractor carries $U = 0.465$; weighting the two attractors by the volume of their basins, the long-run unsafe frequency is discontinuous at $L^{\dagger}$, rising from 0 to 0.32 as L crosses it from below.*

Corollary 16 is the sharpest policy statement available from the model: over an interval spanning a factor of ten in L, more liability means more harm. A regulator moving L upward from a protected regime passes through a range in which the protection it was relying on disappears. It does not recover that protection until L exceeds 42.765, a level at which the external harm of a single unsafe development step is worth more than 42% of the prize for winning the race.

Figure 5 assembles the analytic branches and the two numerical dynamics. The agreement is exact where it should be: the basin-averaged replicator tracks the analytic CS–CAS branch on $(0.116, 0.551)$ and the analytic AS–CAS branch inside the valley, lying slightly below it because a fraction of initial conditions still reaches the safe attractor. Figure 6 shows the corresponding phase portraits; the thick dashed edge visible in every panel is the neutral safe face of Proposition 11, drawn as a continuum of rest points.

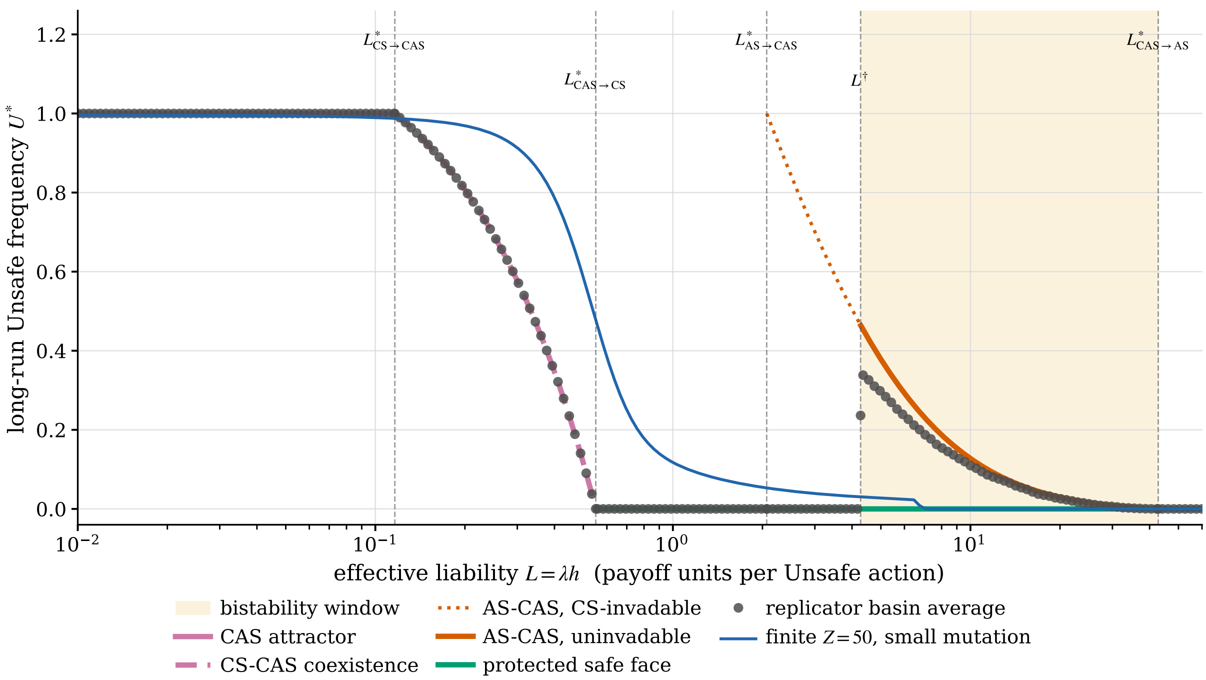

Figure 5: Bifurcation diagram in the effective liability. Thick curves are the analytic branches; markers are the basin-averaged replicator attractor over 110 uniform interior initial conditions; the thin line is the stationary unsafe frequency of the finite-population process ($Z = 50$, small-mutation limit). The shaded band is the bistability window $(L^{\dagger}, L^{*}_{\mathsf{CAS}\to\mathsf{AS}})$ of Corollary 16, on which raising liability raises expected harm.

3.5 Finite populations

The deterministic picture assumes an infinite population and no exploration. Figure 7 reports the stationary value of the observable (7) for the Fermi process over the (L, β) and (L, Z) planes;

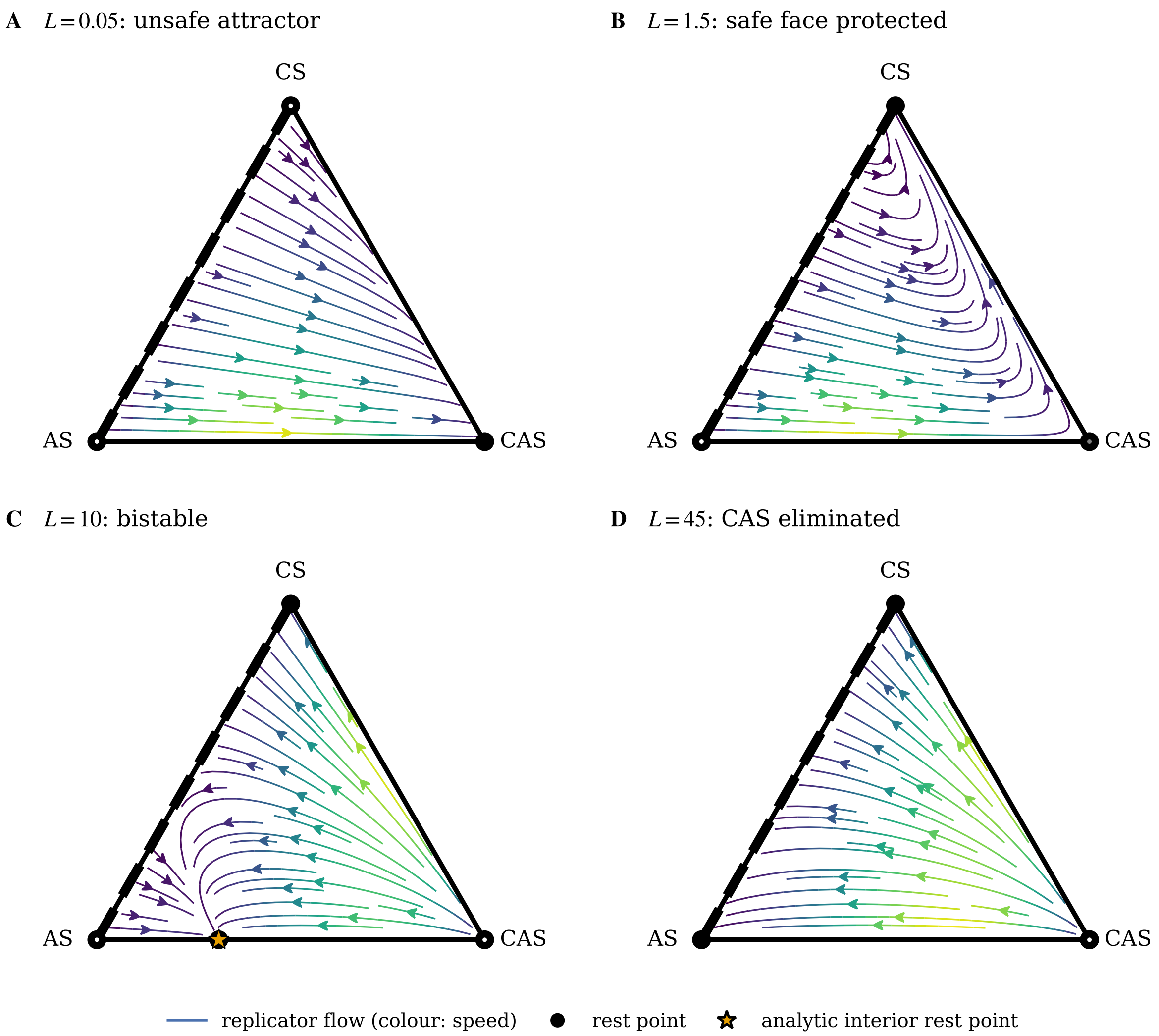

Figure 6: Replicator phase portraits on the $\{\mathsf{AS}, \mathsf{CS}, \mathsf{CAS}\}$ simplex at four liability levels, drawn with the EGTtools simplex utilities [31]. Colour encodes the local speed of selection, scaled within each panel; the thick dashed AS–CS edge is the neutral face. The star marks the analytic interior rest point of Proposition 13. In the valley (C) the flow separates into two basins; at high liability (D) every trajectory reaches the neutral safe face, where it stops at a history-dependent composition.

by Proposition 3 the horizontal axis is again the effective liability alone.

The two thresholds $L^*_{\mathsf{CAS}\to\mathsf{CS}}$ and $L^*_{\mathsf{CAS}\to\mathsf{AS}}$ remain visible as the boundaries of the transition region for all β and Z examined. Weak selection smooths the transition into a broad crossover, as expected: at $\beta \to 0$ imitation is nearly random and the stationary distribution approaches the uniform one over states, so U^* approaches the value implied by a uniform design mixture rather than by any equilibrium. Increasing Z sharpens the transition, consistent with the deterministic limit.

Stochasticity also supplies the transitions between the two attractors of Corollary 16 that the deterministic flow forbids, so bistability becomes a statement about time scales. Table 1 quantifies it at $L = 10$, in the middle of the valley: starting from the unsafe rest point, the mean number of generations needed to reach a state free of CAS ranges from 65 at $Z = 30, \beta = 0.02$ to 3.9×10^{13} at $Z = 100, \beta = 0.2$, and the stationary mass on states in which CAS exceeds a quarter of the population rises from 0.29 to 0.997 over the same range. The valley is therefore not an artefact of the deterministic limit. It is a long-lived trapping region whose residence time grows steeply in both the population size and the intensity of selection, and it is only at weak selection in small populations that liability inside the window buys a protection that is merely slower rather than absent.

Table 1: Escape from the unsafe attractor inside the liability valley ($L = 10$, pool $\{\mathsf{AS}, \mathsf{CS}, \mathsf{CAS}\}$, $\mu = 1/Z$). Mean exit time is the expected number of generations to reach a state free of CAS, starting from the interior rest point of Proposition 13. Unsafe mass is the stationary probability that CAS exceeds a quarter of the population.

Z	β	mean exit (generations)	stationary unsafe mass
30	0.02	6.5×10^{1}	0.291
30	0.20	2.1×10^{4}	0.942
50	0.05	5.9×10^{2}	0.636
80	0.05	5.4×10^{3}	0.768
100	0.05	2.5×10^{4}	0.833
100	0.20	3.9×10^{13}	0.997

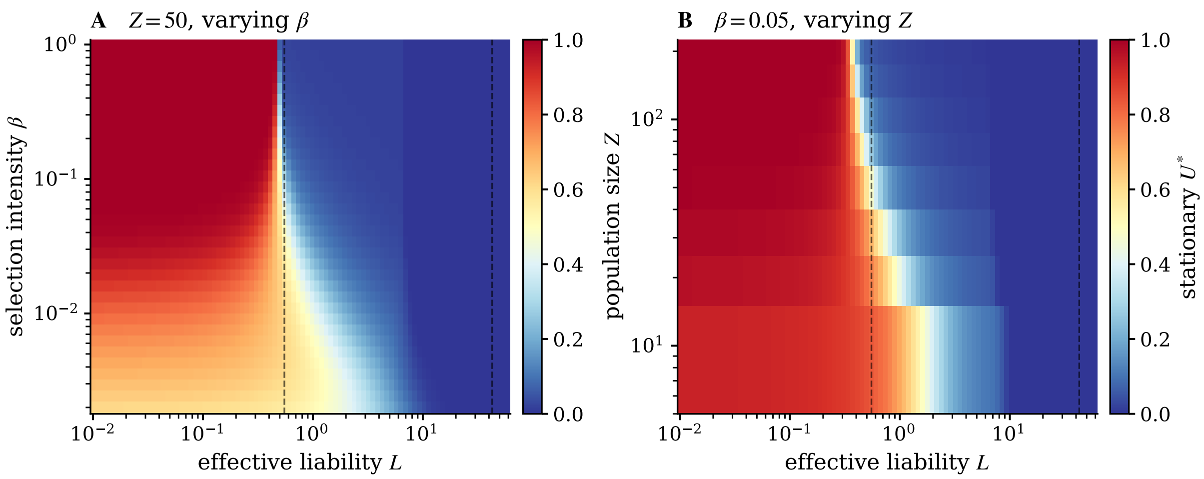

Figure 7: Stationary unsafe frequency of the finite-population pairwise comparison process on $\{\mathsf{AS}, \mathsf{CS}, \mathsf{CAS}\}$ in the small-mutation limit. **(A)** Varying selection intensity at $Z = 50$. **(B)** Varying population size at $\beta = 0.05$. Dashed lines mark $L^*_{\mathsf{CAS}\to\mathsf{CS}}$ and $L^*_{\mathsf{CAS}\to\mathsf{AS}}$.

3.6 Robustness to the constants of the race

Every threshold in (9) is a ratio of payoff differences and therefore inherits the scale of the prize B and the shape of the private risk $p_r^{\max}$. Figure 8 recomputes them over $B \in [10, 400]$ and $p_r^{\max} \in \{0.1, 0.3, 0.6, 0.9\}$. Three findings.

First, the liability that protects unconditional safety is a roughly constant fraction of the prize: $L^*_{\mathsf{CAS}\to\mathsf{AS}}/B$ lies between 0.38 and 0.61 across the whole grid, decreasing weakly in B and in $p_r^{\max}$. The requirement is therefore not an artefact of the particular value $B = 100$; it says that deterring a single unsafe step in a winner-take-all race costs around half the prize, whatever the prize is.

Second, the advantage of conditional over unconditional safety is robust and grows with the prize: at $p_r^{\max} = 0.6$ the ratio rises from 31.7 at $B = 10$ to 90.4 at $B = 400$, and over the whole grid it ranges from 4.4 to 90.4. At $p_r^{\max} = 0.9$ the comparison degenerates in the direction that strengthens the message: $L^*_{\mathsf{CAS}\to\mathsf{CS}}$ becomes negative, meaning that a conditionally safe population repels CAS with *no* liability at all, while unconditional safety still requires $L^*_{\mathsf{CAS}\to\mathsf{AS}} = 38.4$. That CS becomes uninvadable at high private risk also reproduces, from the selection layer, the treatment ordering reported for the interaction layer in the source study [18].

Third, the bistability window exists at every one of the 24 grid points, with multiplicative width between 4.19 and 12.79, widening in both B and $p_r^{\max}$. The liability valley is thus a structural feature of this class of races rather than a coincidence of the baseline parameters.

We also checked the sensitivity of Corollary 10(iii) to the share at which the filtered design is seeded. The gap between the filtered and unfiltered curves stays below 4.2×10^{-3} for seeds up to 20% at every L except $L = 0.3$, where it reaches 1.2×10^{-2} at a 20% seed and 0.12 at a 35% seed. The equivalence is a statement about an ecosystem into which the filtered design enters as a minority, which is the case of interest; it is not claimed for a population already dominated by that design.

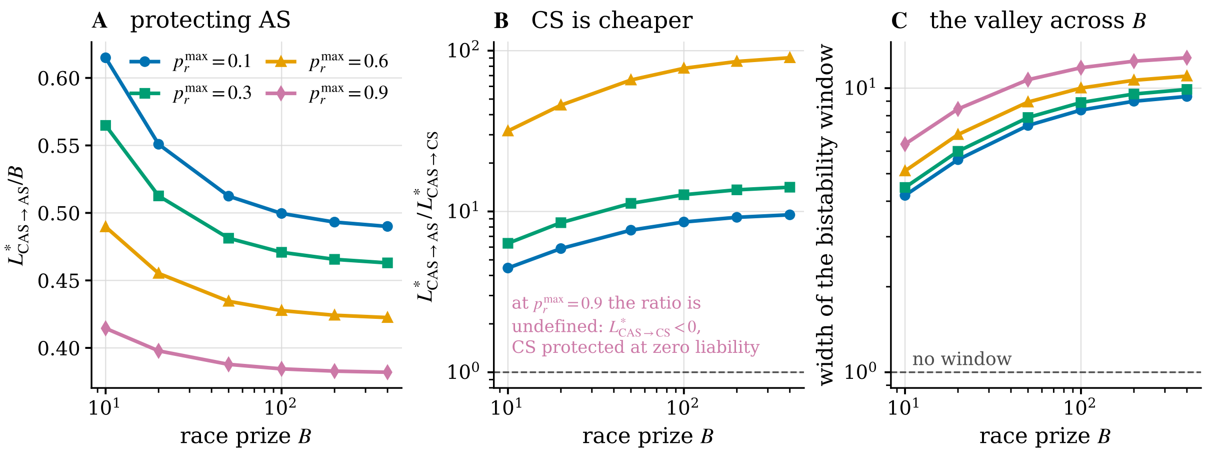

Figure 8: Thresholds across the constants of the race. **(A)** Liability protecting unconditional safety, in units of the prize. **(B)** Ratio $L^*_{\mathsf{CAS}\to\mathsf{AS}}/L^*_{\mathsf{CAS}\to\mathsf{CS}}$, the factor by which unconditional safety is cheaper to invade; the $p_r^{\max} = 0.9$ curve is absent because its denominator is negative, meaning conditional safety needs no liability at all. **(C)** Multiplicative width of the bistability window, which is greater than one at every grid point.

3.7 An eco-evolutionary ratchet

Deployed capability does not disappear. We couple the deployment dynamics to a state $z \in [0, 1]$ measuring the stock of diffused unsafe capability, which enters through enforcement: as a capability proliferates, attributing harm to a particular principal becomes harder, so the liability that actually reaches a principal decays. The coupled system is

$$\dot{x} = \mathrm{Rep}(x;\, a - L_{\mathrm{eff}}(z)\, m), \qquad \dot{z} = \varepsilon\, U(x)\, (1 - z), \tag{15}$$

with $\varepsilon \ll 1$ separating the time scales and L_{eff} an erosion channel. The construction differs from renewable game–environment models [22, 23] in that $\dot{z} \geq 0$ identically: the environmental state is a ratchet.

Theorem 17 (Hysteresis width). *Assume the safe attractor has $U = 0$ exactly, as in Proposition 11, and consider a quasi-static sweep of L that starts in the protected regime. The protected regime is lost at $L_\downarrow = L^*_{CAS\to CS}$, after which $z \to 1$; it is recovered on the return sweep at $L_\uparrow = L^*_{CAS\to CS}/\rho$, where $\rho = L_{\text{eff}}(1)/L$ is the residual fraction of enforcement at full diffusion. In particular*

$$\frac{L_\uparrow}{L_\downarrow} = \frac{1}{\rho} = \begin{cases} \dfrac{1}{1-\theta}, & L_{\text{eff}} = L(1-\theta z), \\ 1+\kappa, & L_{\text{eff}} = L/(1+\kappa z), \end{cases} \tag{16}$$

independently of the diffusion rate ε.

Proof. On the protected branch $U(x) = 0$, so the second equation of (15) gives $\dot z = 0$ and $L_{\text{eff}} = L$; the branch is lost exactly when CAS can invade the resident safe face, i.e. at $L = L^*_{\mathsf{CAS}\to\mathsf{CS}}$ by Lemma 7 and (8). On the unsafe branch $U > 0$, so $z \to 1$ and $L_{\text{eff}} \to \rho L$; recovery requires $L_{\text{eff}} > L^*_{\mathsf{CAS}\to\mathsf{CS}}$, which gives (16). Neither step uses ε. □

The non-trivial ingredient is not the erosion channel but Proposition 11: because the protected branch has $U = 0$ *exactly*, the ratchet variable is frozen there, so the loss threshold is the bare invasion threshold and the width depends on the channel only through the single number ρ. Two different channels with the same residual fraction give the same width, as (16) shows for $\theta = 0.9$ and $\kappa = 9$.

Figure 9 confirms the prediction numerically at $\theta = 0.9$: the descending branch loses protection between $L = 0.62$ and $L = 0.54$, bracketing $L^*_{\mathsf{CAS}\to\mathsf{CS}} = 0.551$, and the ascending branch recovers it at $L = 5.55$ against the predicted 5.51. The loop width is the predicted factor of ten.

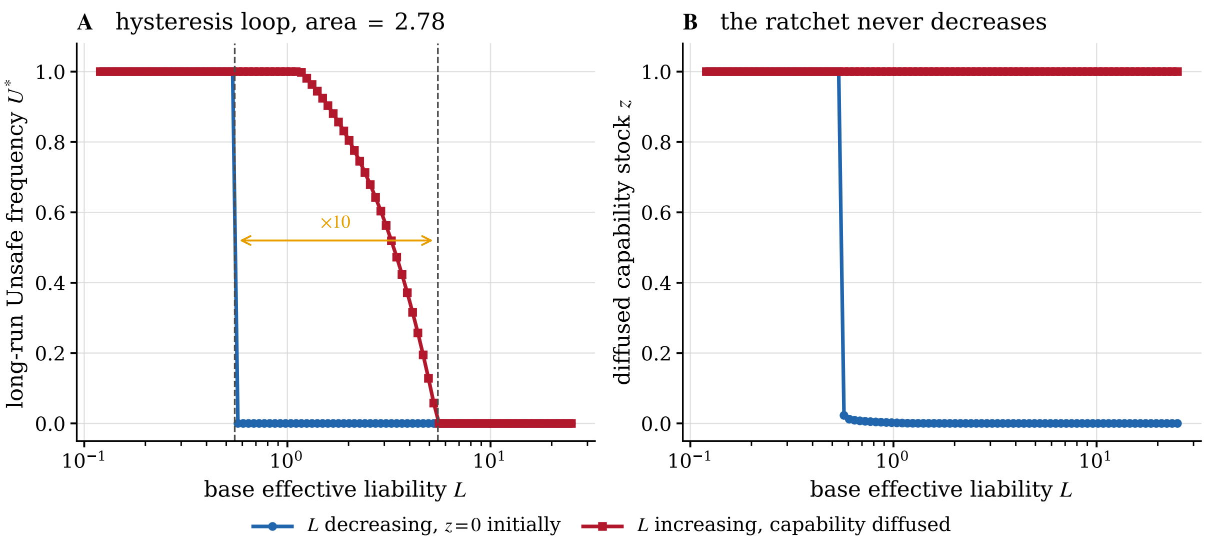

Figure 9: Irreversible capability diffusion produces hysteresis. **(A)** Quasi-static sweep of the base effective liability, decreasing from a protected state and then increasing again. **(B)** The ratchet variable, which never decreases. At $\theta = 0.9$ the recovery threshold is ten times the loss threshold, as Theorem 17 predicts.

The practical content is a timing asymmetry. Because $L_\uparrow$ exceeds $L_\downarrow$ by $1/\rho$, the cost of preventing a transition is smaller than the cost of reversing one by exactly the factor by which enforcement erodes, and the required overshoot is known in closed form once ρ is estimated.

4 Discussion

What the model says and does not say. The results are statements about a specific stylised race with four deterministic designs. They are not predictions about frontier AI systems, and the external harm h has no calibration, which is why every result is stated on the whole L-axis and normalised by the prize. What does transfer is the structure. Whenever the entity that selects a design is not the entity that plays the game, and internalises only part of the harm, the population dynamics is the replicator equation of the selector's functional. The four mechanisms identified here – dominance of conditional over unconditional aggression, neutrality of the safe face, taxation of retaliation, and irreversibility – then follow from qualitative features of the payoff structure rather than from the particular numbers. The first requires only that a conditional design imitate the unconditional one against unsafe opponents. The second requires only that safe designs be behaviourally indistinguishable against safe opponents. The third, by Lemma 14, requires only that the second hold and that retaliation be classified as harmful by the liability rule.

Evaluations measure the wrong object. Corollary 10 says that a filter defined on solo behaviour removes a weakly dominated design. That statement is a formal counterpart of a known practical worry, that a model may behave acceptably when probed and unacceptably in deployment [11, 32]. The mechanism here requires no deception: CAS is not hiding anything. It is simply conditional, and a solo probe cannot elicit the condition. The corrective suggested by the model is not stricter thresholds but a different measurement: score designs by $u(i, i)$, the self-play frequency, or more generally against the environment the design will actually meet. Panel B of Figure 3 is the whole argument – the two orderings disagree on exactly one design, and it is the one that governs the equilibrium.

Do not mandate unconditional safety. Theorem 12 inverts a natural intuition. Unconditional safety looks like the strictest possible design, and it is the most fragile: it demands the highest liability of any safe composition, by a factor of 77.7 at the baseline and up to 90.4 at large prizes. Conditional safety is cheap to protect because it makes exploiting it unprofitable, and at high private risk it needs no liability at all. Because the safe face is neutral, a policy that fixes the composition is not self-sustaining; it must be maintained against drift.

Liability has a valley, and it is wide. Corollary 16 is the result most likely to matter outside the model. Liability instruments are usually assumed to be monotone: more internalisation, less harm. Here an entire decade of L is worse than the lower end of the range, because the instrument does not distinguish unsafe behaviour that initiates from unsafe behaviour that retaliates. Theorem 15 locates the boundary exactly, and Section 3.6 shows that a window of this kind exists across the whole range of prizes and risk levels we examined. The distinction between initiating and retaliating is available in principle – it is a matter of conditioning the rule on the counterparty's prior action, which is exactly the column-constant structure that Proposition 6 identifies as inert. Designing an implementable approximation of that rule is the natural next question.

Structured populations. The analysis is well mixed. Network reciprocity generally favours conditionally cooperative strategies by clustering them [33, 34], and we expect it to act unevenly on our four mechanisms. The weak dominance of CAS over AU (Theorem 9) is a pointwise payoff comparison and survives any matching structure. The neutrality of the AS–CS face (Proposition 11) is likewise a statement about identical payoffs and survives, although the drift dynamics on it does not. By contrast the valley depends on the balance between two invasion conditions and should narrow under assortment, since clustering raises the payoff a conditionally safe design obtains from its own kind without raising the retaliation it must pay for. Establishing whether it closes is left open.

Limitations. Four are material. The interaction layer has two players and a fixed pool of four designs, so entry of new designs is modelled only as mutation. The principals are homogeneous and passive: they select on π_P but do not strategise about each other, which rules out the enforcement-avoidance behaviour a richer model would include. The ratchet channel is assumed rather than derived; Theorem 17 isolates what does not depend on that assumption, but the assumption of erosion itself remains. Finally, the analysis is deterministic in h and λ, whereas real liability arrives with probability and delay, so an expected-value reading of L is a first approximation only.

4.1 Conclusion

Separating the payoff that an AI design earns from the payoff that decides whether it is replicated changes the dynamical system, not merely its parameters. Working through that separation in the reduced AI race yields four results that are invisible when the two payoffs are identified. Pre-deployment solo evaluation cannot change the equilibrium, because it removes a weakly dominated design. The invasion barrier protecting a safe population is controlled by a neutrally drifting composition, and falls by a factor of 77.7 along it. Long-run harm is non-monotone in liability, with a bistable window that exists at every prize and risk level we examined and traps the population for up to 4×10^{13} generations. And irreversible capability diffusion makes the loss of a protected regime cost a known factor less than its recovery.

Each result points at the same target. Governing an ecosystem of deployed agents by acting on the designs – filtering them, training them, mandating them – addresses the layer that does not carry the selection gradient. Acting on the functional that principals select by does carry it, but the instrument is not monotone, and the range in which it backfires is precisely the range in which it is strong enough to punish retaliation and too weak to protect unconditional safety. Locating that range for a given deployment setting seems to us more useful than arguing about whether liability should be strengthened.

Two extensions follow directly. Estimating the payoff and harm matrices of real agent designs empirically, rather than from a stylised race, would place an actual deployment ecosystem on the L-axis. And enriching the principals – letting them anticipate liability, compete, or share it – turns the selection functional itself into a strategic object, which is the natural second layer of the problem.

Declarations

Code and data availability. All code, tables and figures are reproducible from a public repository released under the MIT licence, https://github.com/trungkiet2005/deployment-layer-selection. Evolutionary computations use EGTtools [31].

Competing interests. The authors declare no competing interests.

Funding. To be completed.

A Matchup tables and thresholds

Table 2: Task payoffs $a(i, j)$ of the reduced race game at $p_r^{\max} = 0.6$, computed exactly over the horizon distribution. Rows index the focal design.

	AS	AU	CS	CAS
AS	59.000	5.400	59.000	8.600
AU	48.640	27.200	47.360	27.200
CS	59.000	16.600	59.000	26.644
CAS	101.765	27.200	61.631	27.200

Table 3: Expected number of Unsafe actions $m(i, j)$ of the focal design at $p_r^{\max} = 0.6$.

	AS	AU	CS	CAS
AS	0.000	0.000	0.000	0.000
AU	9.000	9.000	9.000	9.000
CS	0.000	8.000	0.000	4.222
CAS	1.000	9.000	4.778	9.000

Table 4: Closed-form critical effective liabilities $L^* = \delta a/\delta m$ for every ordered invasion at $p_r^{\max} = 0.6$. The last column states when the invader has a positive growth rate.

Invader	Resident	δa	δm	L^*	invades
AS	AU	-21.800	-9.000	2.422	$L > L^*$
AS	CS	0.000	0.000	–	–
AS	CAS	-18.600	-9.000	2.067	$L > L^*$
AU	AS	-10.360	9.000	-1.151	$L < L^*$
AU	CS	-11.640	9.000	-1.293	$L < L^*$
AU	CAS	0.000	0.000	–	–
CS	AS	0.000	0.000	–	–
CS	AU	-10.600	-1.000	10.600	$L > L^*$
CS	CAS	-0.556	-4.778	0.116	$L > L^*$
CAS	AS	42.765	1.000	42.765	$L < L^*$
CAS	AU	0.000	0.000	–	–
CAS	CS	2.631	4.778	0.551	$L < L^*$

B The other risk treatments

The source experiment manipulates the maximum private setback risk $p_r^{\max} \in \{0.1, 0.6, 0.9\}$ in (2). The unsafe counts $m(i,j)$ are identical across treatments, because $p_r^{\max}$ scales the realised payoff but leaves the stage matrix (1) and hence the action paths untouched; only $a(i,j)$ changes. Evaluating the interaction game without liability ($L = 0$) reproduces the qualitative ordering reported in the source study [18]: the neutrally stable set is $\{\mathsf{AU}, \mathsf{CAS}\}$ at $p_r^{\max} = 0.1$ and 0.6, and $\{\mathsf{CS}\}$ at $p_r^{\max} = 0.9$, so high private risk favours conditional safety even before any external liability is applied. Our main analysis uses $p_r^{\max} = 0.6$, the intermediate treatment, because it is the one in which the conditional designs are both present and neither is trivially dominant; Section 3.6 reports every threshold for the other treatments.

References

[1] John Maynard Smith. *Evolution and the Theory of Games.* Cambridge University Press, 1982.

[2] Peter D. Taylor and Leo B. Jonker. Evolutionary stable strategies and game dynamics. *Mathematical Biosciences*, 40(1–2):145–156, 1978. doi: 10.1016/0025-5564(78)90077-9.

[3] Josef Hofbauer and Karl Sigmund. *Evolutionary Games and Population Dynamics.* Cambridge University Press, 1998.

[4] Garrett Hardin. The tragedy of the commons. *Science*, 162(3859):1243–1248, 1968. doi: 10.1126/science.162.3859.1243.

[5] Werner Güth and Menahem E. Yaari. An evolutionary approach to explain reciprocal behavior in a simple strategic game. In Ulrich Witt, editor, *Explaining Process and Change: Approaches to Evolutionary Economics*, pages 23–34. University of Michigan Press, 1992.

[6] Werner Güth. An evolutionary approach to explaining cooperative behavior by reciprocal incentives. *International Journal of Game Theory*, 24(4):323–344, 1995. doi: 10.1007/BF01243036.

[7] Eddie Dekel, Jeffrey C. Ely, and Okan Yilankaya. Evolution of preferences. *The Review of Economic Studies*, 74(3):685–704, 2007. doi: 10.1111/j.1467-937X.2007.00445.x.

[8] Ingela Alger and Jörgen W. Weibull. Homo moralis—preference evolution under incomplete information and assortative matching. *Econometrica*, 81(6):2269–2302, 2013. doi: 10.3982/ECTA10637.

[9] Aviad Heifetz, Chris Shannon, and Yossi Spiegel. What to maximize if you must. *Journal of Economic Theory*, 133(1):31–57, 2007. doi: 10.1016/j.jet.2005.09.008.

[10] Dan Hendrycks. Natural selection favors AIs over humans. *arXiv preprint arXiv:2303.16200*, 2023.

[11] Lewis Hammond et al. Multi-agent risks from advanced AI. Technical Report Technical Report #1, Cooperative AI Foundation, 2025. arXiv:2502.14143.

[12] Stuart Armstrong, Nick Bostrom, and Carl Shulman. Racing to the precipice: a model of artificial intelligence development. *AI & Society*, 31(2):201–206, 2016. doi: 10.1007/s00146-015-0590-y.

[13] The Anh Han, Luís Moniz Pereira, Francisco C. Santos, and Tom Lenaerts. To regulate or not: A social dynamics analysis of an idealised AI race. *Journal of Artificial Intelligence Research*, 69:881–921, 2020. doi: 10.1613/jair.1.12225.

[14] Theodor Cimpeanu, Francisco C. Santos, Luís Moniz Pereira, Tom Lenaerts, and The Anh Han. Artificial intelligence development races in heterogeneous settings. *Scientific Reports*, 12:1723, 2022. doi: 10.1038/s41598-022-05729-3.

[15] Amanda Askell, Miles Brundage, and Gillian K. Hadfield. The role of cooperation in responsible AI development. *arXiv preprint arXiv:1907.04534*, 2019.

[16] Elias Fernández Domingos, Inês Terrucha, Rémi Suchon, Jelena Grujic, Juan C. Burguillo, Francisco C. Santos, and Tom Lenaerts. Delegation to artificial agents fosters prosocial behaviors in the collective risk dilemma. *Scientific Reports*, 12:8492, 2022. doi: 10.1038/s41598-022-11518-9.

[17] Inês Terrucha, Elias Fernández Domingos, Manuel Simão, Tom Lenaerts, and Francisco C. Santos. Committing to the wrong artificial delegate in a collective-risk dilemma is better than directly committing mistakes. *Scientific Reports*, 14:10460, 2024. doi: 10.1038/s41598-024-61153-9.

[18] Elias Fernández Domingos and The Anh Han. Falling behind drives unsafe development in an idealised AI race experiment. *arXiv preprint arXiv:2607.26034*, 2026.

[19] Robert Axelrod and William D. Hamilton. The evolution of cooperation. *Science*, 211 (4489):1390–1396, 1981. doi: 10.1126/science.7466396.

[20] Martin Nowak and Karl Sigmund. A strategy of win-stay, lose-shift that outperforms tit-for-tat in the Prisoner's Dilemma game. *Nature*, 364:56–58, 1993. doi: 10.1038/364056a0.

[21] Pedro Dal Bó and Guillaume R. Fréchette. On the determinants of cooperation in infinitely repeated games: A survey. *Journal of Economic Literature*, 56(1):60–114, 2018. doi: 10.1257/jel.20160980.

[22] Joshua S. Weitz, Ceyhun Eksin, Keith Paarporn, Sam P. Brown, and William C. Ratcliff. An oscillating tragedy of the commons in replicator dynamics with game-environment feedback. *Proceedings of the National Academy of Sciences*, 113(47):E7518–E7525, 2016. doi: 10.1073/pnas.1604096113.

[23] Andrew R. Tilman, Joshua B. Plotkin, and Erol Akçay. Evolutionary games with environmental feedbacks. *Nature Communications*, 11:915, 2020. doi: 10.1038/s41467-020-14531-6.

[24] Marten Scheffer, Steve Carpenter, Jonathan A. Foley, Carl Folke, and Brian Walker. Catastrophic shifts in ecosystems. *Nature*, 413:591–596, 2001. doi: 10.1038/35098000.

[25] Steven Shavell. Strict liability versus negligence. *The Journal of Legal Studies*, 9(1):1–25, 1980. doi: 10.1086/467626.

[26] Steven Shavell. A model of the optimal use of liability and safety regulation. *The RAND Journal of Economics*, 15(2):271–280, 1984. doi: 10.2307/2555680.

[27] Karthik Panchanathan and Robert Boyd. Indirect reciprocity can stabilize cooperation without the second-order free rider problem. *Nature*, 432:499–502, 2004. doi: 10.1038/nature02978.

[28] Karl Sigmund, Hannelore De Silva, Arne Traulsen, and Christoph Hauert. Social learning promotes institutions for governing the commons. *Nature*, 466:861–863, 2010. doi: 10.1038/nature09203.

[29] Arne Traulsen, Martin A. Nowak, and Jorge M. Pacheco. Stochastic dynamics of invasion and fixation. *Physical Review E*, 74:011909, 2006. doi: 10.1103/PhysRevE.74.011909.

[30] Drew Fudenberg and Lorens A. Imhof. Imitation processes with small mutations. *Journal of Economic Theory*, 131(1):251–262, 2006. doi: 10.1016/j.jet.2005.04.006.

[31] Elias Fernández Domingos, Francisco C. Santos, and Tom Lenaerts. EGTtools: Evolutionary game dynamics in Python. *iScience*, 26(4):106419, 2023. doi: 10.1016/j.isci.2023.106419.

[32] Teun van der Weij, Felix Hofstätter, Ollie Jaffe, Samuel F. Brown, and Francis Rhys Ward. AI sandbagging: Language models can strategically underperform on evaluations. *arXiv preprint arXiv:2406.07358*, 2024.

[33] Hisashi Ohtsuki, Christoph Hauert, Erez Lieberman, and Martin A. Nowak. A simple rule for the evolution of cooperation on graphs and social networks. *Nature*, 441:502–505, 2006. doi: 10.1038/nature04605.

[34] Francisco C. Santos, Jorge M. Pacheco, and Tom Lenaerts. Evolutionary dynamics of social dilemmas in structured heterogeneous populations. *Proceedings of the National Academy of Sciences*, 103(9):3490–3494, 2006. doi: 10.1073/pnas.0508201103.